

Network Science — Chapter 3: Random Networks

Formula Sheet — All Symbols Defined

Barabási (2016)

1 MODEL SETUP

N nodes; each pair connected independently with probability p .

N = total number of nodes

p = connection probability, $0 \leq p \leq 1$

$L_{\max} = \frac{N(N-1)}{2}$, max possible links

2 LINKS & AVERAGE DEGREE

$$\langle L \rangle = p \cdot \frac{N(N-1)}{2}$$

$\langle L \rangle$ = expected number of links

p = connection probability

$\frac{N(N-1)}{2}$ = total possible pairs

$$\langle k \rangle = p(N-1)$$

$\langle k \rangle$ = average degree (mean links per node)

$N-1$ = max degree (links to all other nodes)

3 DEGREE DISTRIBUTION

Exact — Binomial (valid for any N):

$$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

p_k = prob. a node has degree k

$\binom{N-1}{k}$ = ways to choose k from $N-1$ links

p^k = prob. those k links exist

$(1-p)^{N-1-k}$ = prob. remaining links absent

Poisson approximation (when $\langle k \rangle \ll N$):

$$p_k = \frac{e^{-\langle k \rangle} \langle k \rangle^k}{k!}$$

$e^{-\langle k \rangle}$ = normalisation factor

$\langle k \rangle^k$ = grows with mean degree

$k!$ = suppresses high- k ; drops faster than exp.

$\langle k \rangle$ = **sole parameter** (no dependence on N !)

Poisson moments:

$$\sigma_k = \sqrt{\langle k \rangle}, \quad \langle k^2 \rangle = \langle k \rangle (\langle k \rangle + 1)$$

σ_k = std. deviation of degree

$\sigma_k / \langle k \rangle = \langle k \rangle^{-1/2} \rightarrow 0$ for large $\langle k \rangle$

Hub suppression (Stirling approx.):

$$p_k \approx \frac{e^{-\langle k \rangle}}{\sqrt{2\pi k}} \left(\frac{e\langle k \rangle}{k} \right)^k$$

For $k > e\langle k \rangle$: the bracket < 1 , so p_k drops **faster than exponentially**. Hubs are virtually impossible.

4 PHASE TRANSITIONS & GIANT COMPONENT

Giant component (GC) threshold:

$$\langle k \rangle = 1 \iff p_c = \frac{1}{N-1} \approx \frac{1}{N}$$

$\langle k \rangle = 1$ = min. avg. degree for GC to exist

p_c = critical connection probability

GC size (supercritical, near threshold):

$$\frac{N_G}{N} \sim \langle k \rangle - 1, \quad N_G \sim (p - p_c)N$$

N_G = number of nodes in the giant component

$\langle k \rangle - 1$ = how far above the threshold $\langle k \rangle = 1$

$p - p_c$ = excess prob. above critical value

Full connectivity threshold:

$$\langle k \rangle = \ln N \iff p = \frac{\ln N}{N}$$

$\ln N$ = avg. degree at which *all* nodes join GC

Four regimes:

Regime	Condition	N_G scaling
Subcritical	$\langle k \rangle < 1$	$\ln N$; $N_G/N \rightarrow 0$
Critical	$\langle k \rangle = 1$	$N^{2/3}$; $N_G/N \rightarrow 0$
Supercritical	$1 < \langle k \rangle < \ln N$	$N_G/N \sim \langle k \rangle - 1$
Connected	$\langle k \rangle > \ln N$	$N_G \approx N$

5 SMALL WORLDS — PATH LENGTH

Reachable nodes within d hops:

$$N(d) \approx \langle k \rangle^d$$

$N(d)$ = nodes reachable within d steps

$\langle k \rangle^d$ = grows exponentially with d

Average path length (small-world formula):

$$\langle d \rangle \approx \frac{\ln N}{\ln \langle k \rangle}$$

$\langle d \rangle$ = avg. shortest-path distance

$\ln N$ = grows very slowly — paths stay short

$\ln \langle k \rangle$ = larger $\langle k \rangle \Rightarrow$ shorter paths

Lattice comparison ($\langle d \rangle$ scaling with N):

Network type	$\langle d \rangle$ scales as
Random network	$\ln N$ (small world!)
1D lattice (line)	N
2D lattice (grid)	$N^{1/2}$
d -dim. lattice	$N^{1/d}$

Social network: $N \approx 7 \times 10^9$, $\langle k \rangle \approx 10^3$

$$\langle d \rangle = \frac{\ln(7 \times 10^9)}{\ln(10^3)} \approx 3.3$$

Milgram (1967): 5.2 — Facebook (2011): 4.74

WWW (finite-size scaling):

$$\langle d \rangle_{\text{WWW}} \approx 0.35 + 0.89 \ln N$$

At $N \approx 8 \times 10^8$: $\langle d \rangle \approx 18.7$ (“19 degrees”)

6 CLUSTERING COEFFICIENT

Local clustering coefficient:

$$C_i = \frac{2L_i}{k_i(k_i - 1)}$$

L_i = actual links among node i 's neighbours

$k_i(k_i - 1)/2$ = max possible links among k_i neighbours

$C_i = 0$ = no neighbours are connected

$C_i = 1$ = all neighbours fully interconnected

In a random network:

$$C_i = p = \frac{\langle k \rangle}{N}$$

C_i = same for *every* node regardless of k_i

$\langle k \rangle / N$ = decreases as $1/N$ (tiny for large networks)

Failures vs. real networks:

- Real $\langle C \rangle$ is *independent* of N (not $\propto 1/N$)
- Real $C(k)$ *decreases* with k (not constant)
- Real $\langle C \rangle \gg \langle k \rangle / N$ (far higher clustering)

7 BINOMIAL DISTRIBUTION (reference)

$$P(X=x) = \binom{N}{x} p^x (1-p)^{N-x}$$

$$\langle x \rangle = Np, \quad \sigma_x = \sqrt{Np(1-p)}$$

$$\langle x^2 \rangle = p(1-p)N + p^2 N^2$$

$\langle x \rangle$ = mean (first moment)

σ_x = standard deviation

$\langle x^2 \rangle$ = second moment

In the Barrat weighted clustering definition:

$$C_i^w = \frac{1}{s_i(k_i - 1)} \sum_{j,h} \frac{w_{ij} + w_{ih}}{2} a_{ij} a_{ih} a_{jh}$$

COMPLETE FORMULA REFERENCE TABLE

#	Quantity	Formula	Symbol meanings
1	Max. pairs	$L_{\max} = \frac{N(N-1)}{2}$	N : total nodes
2	Expected links	$\langle L \rangle = p \frac{N(N-1)}{2}$	p : connection probability
3	Average degree	$\langle k \rangle = p(N-1)$	$N-1$: max degree of any node
4	Degree dist. (binomial)	$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}$	exact form, valid for all N
5	Degree dist. (Poisson)	$p_k = \frac{e^{-\langle k \rangle} \langle k \rangle^k}{k!}$	approx., valid when $\langle k \rangle \ll N$
6	Degree std. deviation	$\sigma_k = \sqrt{\langle k \rangle}$	width of degree distribution
7	Second moment	$\langle k^2 \rangle = \langle k \rangle (\langle k \rangle + 1)$	used in percolation theory
8	Critical probability	$p_c = \frac{1}{N-1} \approx \frac{1}{N}$	threshold for giant component
9	GC condition	$\langle k \rangle > 1$	necessary & sufficient for GC
10	GC size (supercritical)	$\frac{N_G}{N} \sim \langle k \rangle - 1$	N_G : nodes in giant component
11	GC size (equivalent)	$N_G \sim (p - p_c)N$	linear growth above threshold
12	Full connectivity	$\langle k \rangle > \ln N$	all nodes absorbed into GC
13	Nodes within d hops	$N(d) \approx \langle k \rangle^d$	exponential growth with distance
14	Average path length	$\langle d \rangle \approx \frac{\ln N}{\ln \langle k \rangle}$	logarithmic in N — small world!
15	Clustering coefficient	$C_i = p = \frac{\langle k \rangle}{N}$	same for all nodes; $\rightarrow 0$ as $N \rightarrow \infty$